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COMPARATIVE STUDY OF FOUNDATIONAL TIME SERIES MODELS FOR EMERGENCY DEPARTMENT OCCUPANCY FORECASTING

Bachelor's Thesis
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ABSTRACT

Valtti Järvi: Comparative study of foundational time series models for emergency department occupancy forecasting

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Emergency department overcrowding is a significant threat to patient safety..

Keywords: time series forecasting, foundational models, emergency department occupancy prediction

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TIIVISTELMÄ

Valtti Järvi:
Kandidaatin tutkielma
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Tieto- ja sähkötekniikan kandidaattiohjelma
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Avainsanat: keyword, keyword, keyword, keyword, keyword

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck -ohjelmalla.

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1. INTRODUCTION

Overcrowding in hospital emergency departments (EDs) have a clear negative impact on patients health [1]. For example, a higher mortality rate [2][3] and increased length of stay [4] have been associated with crowding. Accurate predictions of future ED occupancy levels could prevent crowding by allocating resources differently.

Current state of the art (SOTA) models for predicting ED occupancy are trained by using historical occupancy data and sometimes other factors, like weather and traffic data. (lähde) Training machine learning models is computationally heavy and time consuming task. Every hospital having unique characteristics, model trained with data from one hospital could possibly not predict well future occupancy levels for other hospitals. Foundational models do not need training to predict from specific data. If they could predict ED occupancy accurately enough, hard part of model training can be skipped. This could save substantial amount of computation and man hours.

In this thesis foundational time series models are evaluated using ED occupancy data from Acuta. Acuta is the name of Tampere university hospitals (TAYS) emergency department. TAYS is located in Tampere, Finland. Data used in this thesis is identical to the data Tuominen et al. [5] used in their study. Tuominen et al. created machine learning models to predict ED occupancy. Since the data is identical, direct comparison between foundational models and models from study by Tuominen et al. is possible. Foundational model performance can then be linked to other trained machine learning models.

In this thesis foundational time series models are evaluated using ED occupancy data from Acuta. Acuta is the name of Tampere university hospitals (TAYS) emergency department. TAYS is located in Tampere, Finland. The data used is identical to that in a study by Tuominen et al. [5]. In their work, they created machine learning models to predict ED occupancy. This allows for a direct comparison between the foundational models and their results.

In this thesis foundational time series models are evaluated using ED occupancy data from Acuta. Acuta is the name of Tampere university hospitals (TAYS) emergency department. TAYS is located in Tampere, Finland. The dataset is identical to the one used by Tuominen et al. [5] to train machine learning models for occupancy prediction. Because the data is the same, a direct performance comparison between foundational models and

their models is possible.

The goal of this thesis was to examine how well foundational time series models perform on emergency department occupancy prediction.

Five different foundational time series models were tested against trained machine learning models using same validation data. Different context lengths were tested. Giving multivariate data as a input was also tested.

2. BACKGROUND

2.1 Time series forecasting in healthcare

2.1.1 Motivation

- why is forecasting in healthcare important? (improve patient safety, resource allocation...)

2.1.2 ED time series data

- what is time series data
- how to forecasting is done (looking at historical data, capturing cyclical patterns)
- erityispiirteitä ED aikasarjadatasta. Mikä on occupancy, miksi sitä ennustetaan eikä arrivals, miten ED data käyttäytyy lyhyellä aikavälillä

2.2 Foundational models

2.2.1 Definiton

- what foundational models are (trained using massive amounts of data etc)

2.2.2 Models

- short and high level description of architectures and features of each model. naming of each model
- very short description of trained ml models that these models are compared against

2.2.3 Paradigm shift

- paradigm from ml model training+validation -> foundational models (saves computation, time and effort, is generalizeble)

2.3 Related work

2.3.1 Current state of the art in ED forecasting

- tietoa nykyisestä state of artista ED ennustuksessa

2.3.2 Performance of foundational models in other fields

- Performance of foundational time series prediction for ED occ in other studies (if any)
- Performance of foundational time series prediction for other data (find studies)

3. METHODOLOGY

3.1 Models

- criteria for choosing models
- settings and hyper-parameters that were used for each model
- very short description of trained ml models that these models are compared against

3.2 Data

- where the data came from
- describe the data used (univariate, multivariate...) and how it was processed
- data allows direct comparison with other ml models trained on the same data.

3.3 Experimental setup

- task is to predict 24h occupancy to the future
- uni vs multivariate inputs
- different context length inputs
- hardware setup and (library versions, date when models were downloaded?)

3.4 Evaluation

- describe the evaluation metrics used (rmse, mae, msis), what they tell and why they are used

3.5 Finetuning

- tämä osio jos finetunausta tehdään

4. RESULTS

4.1 Main results

- show the charts and tables of the main results (rmse, mae, msis)

4.2 Effect of context length

- charts that show error with different context lengths

4.3 Effect of covariates

- chart comparing uni vs multi inputs

4.4 Visualizations

- show single day prediction vs ground truth. (best, worst and typical days)

5. DISCUSSION

- how accuracy changes with input length
- how input covariates changes accuracy
- did fine tuning increase accuracy
- are foundational models potential option for areas like this one, comparison to trained models, compare with other studies: Show chart with all the old results and new results. Compare them and keep telling the story of trained vs foundational
- limitations of the study (only one validation dataset)
- other observations and future work
- maybe connect with results or conclusion chapter?

6. CONCLUSION

- summary of answers to the research questions, conclusion

7. REFERENCES

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